State-Aware Desk Lighting Control with On-Device Image Classification on XIAO ESP32S3 Sense

Yian Fan
Columbia University
yf2768@columbia.edu

Chen Gao
Columbia University
cg3607@columbia.edu

Abstract

This report presents a camera-based embedded AI desk lighting controller built on the Seeed Studio XIAO ESP32S3 Sense. The system uses the onboard camera to classify a fixed desk scene into four states: computer use, empty desk, iPad use, and phone use. A lightweight MobileNetV2-based image classification model was trained in Edge Impulse using images captured directly from the XIAO camera and deployed to the board as an Arduino library for on-device inference. The predicted class is mapped to a lighting behavior and used to control a two-channel cool-white/warm-white LED strip. The XIAO generates two independent pulse-width modulation (PWM) signals through pins D1 and D0, which drive two D4184 MOSFET switching channels connected to a 12 V LED load. The final design simplifies the original broader user-state recognition idea into a fixed-view desk scene classification task, which is more reliable for a small embedded model, limited image quality, and a small project dataset.

1 Introduction

Manual indoor lighting control is simple but inefficient when the user's activity changes frequently. Occupancy-based lighting systems can reduce unnecessary energy use, but simple presence sensing cannot distinguish between different desk activities or lighting needs [18, 19]. A desk environment may require different lighting conditions depending on whether the user is working on a computer, reading or using an iPad, using a phone, or away from the desk.

The original project proposal considered a broader state-monitoring system using camera and microphone inputs to recognize user states such as working, idle, or sleeping [1]. During implementation, we revised the design to a camera-only fixed-desk scene classification system on the XIAO ESP32S3 Sense. This change made the project more feasible for the available hardware, dataset size, and deployment time. The XIAO ESP32S3 Sense provides a compact ESP32S3-based platform with camera support, PSRAM, flash storage, and embedded ML compatibility [2, 4].

The final prototype demonstrates a complete sensing-inference-control pipeline. The XIAO camera captures a local image, a MobileNetV2-based classifier trained in Edge Impulse predicts the desk state, and firmware maps the predicted class to PWM duty cycles for the cool-white and warm-white LED channels. The system runs inference on-device and does not require cloud processing.

2 Task Definition and System Architecture

The final recognition task is a four-class fixed-view desk scene classification problem. Instead of detecting exact hand positions or object bounding boxes, the model classifies the whole camera frame into one of four labels: *computer*, *empty*, *ipad*, and *phone*. These labels were selected because they directly correspond to lighting behaviors. The original idea of detecting hand-object interactions was difficult because the XIAO camera image was low-resolution and noisy, hands often overlapped with objects, and a hand could rest near the keyboard or mouse without actually indicating computer use.

Figure 1 shows the high-level system architecture. A camera frame is captured by the XIAO ESP32S3 Sense and processed by our MobileNetV2-based desk classifier, trained in Edge Impulse and deployed as an Arduino library. The firmware selects the highest-confidence class and applies a rule-based lighting map. The selected mode is converted into two PWM duty cycles for the cool-white and warm-white LED channels.

The architecture separates the low-power logic domain and the higher-power lighting domain. The XIAO operates at 3.3 V logic level, while the LED strip uses an external 12 V supply. The D4184 MOSFET modules bridge these two domains by accepting PWM control signals and switching the LED current. This avoids driving the LED strip directly from microcontroller GPIO pins.

3 Model, Data, and Iteration

The image classifier was trained using Edge Impulse. Edge Impulse provides an image classification workflow for collecting images, applying transfer learning, and deploying trained impulses to embedded devices [5]. To reduce the domain gap between training and deployment, the dataset was collected directly from the XIAO ESP32S3 Sense camera instead of using a phone camera or external webcam. During data collection, the XIAO ran a temporary HTTP capture server that returned a JPEG image from the endpoint `/capture`. A Python script on a Windows laptop saved the captured images into label-specific folders. This also avoided the need for a working microSD card.

The final dataset used four labels: *computer*, *empty*, *ipad*, and *phone*. The camera was fixed at the same desk angle used during deployment. The *empty* class included hard negative examples such as a desk with keyboard, mouse, phone, or iPad visible but without active user interaction. These examples helped reduce false activation when objects were present but the user was away.

The final model was trained using transfer learning with a lightweight MobileNetV2 96×96 0.35 image classification model. MobileNetV2 is designed for mobile and resource-constrained vision tasks using inverted residual blocks and linear bottlenecks [7]. The original camera frames were captured at QVGA resolution, and Edge Impulse resized the images to the model input size. After training, the model was exported as an Arduino library and integrated into the XIAO firmware. Edge Impulse Arduino deployment packages the processing blocks, configuration, and learning block into a library that can be included in a sketch for local inference [6].

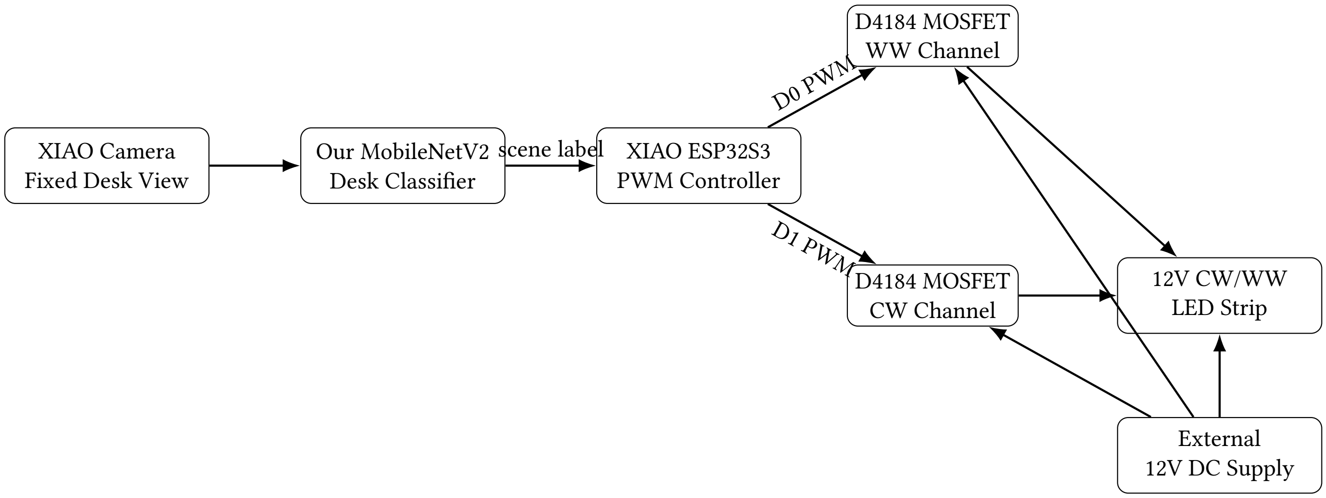

Figure 1: Overall architecture of the camera-based desk lighting controller. The XIAO ESP32S3 Sense runs image classification locally and uses PWM signals to control the MOSFET-driven LED channels.

The final model achieved 93.8% validation accuracy with a validation loss of 0.22. The validation confusion matrix showed perfect classification for the *computer*, *empty*, and *phone* classes, while the *ipad* class was the most difficult. In the validation set, 81.8% of *ipad* samples were correctly classified, while 18.2% were confused with *phone*. The weighted average precision, recall, and F1 score were 0.95, 0.94, and 0.94, respectively. This was acceptable for the prototype because the main safety-critical distinction was whether the desk was empty or active, and the *empty* class was correctly separated in validation.

Several alternative model choices were considered. MobileNetV1 is also designed for efficient mobile vision and uses depthwise separable convolutions with width and resolution multipliers [8]. A smaller MobileNetV1 or MobileNetV2 variant could reduce latency, but it may have lower capacity for visually similar classes such as iPad and phone use. FOMO was also considered because it provides lightweight object detection for constrained devices [9]. However, it would require object-level labels and was less suitable for our low-resolution desk view where hands and objects frequently overlap. Larger classifiers such as EfficientNet may improve accuracy but increase model complexity [10]; compact architectures such as SqueezeNet reduce parameter count but are not as directly supported in our Edge Impulse Arduino workflow [11]. Therefore, MobileNetV2 96×96 0.35 was selected as a practical balance between transfer-learning accuracy, model size, and deployment compatibility.

Before the final four-class model, we tried a broader three-class scene model with the labels *computer use*, *empty desk*, and *reading*. Each class contained around 170 captured images. Although the training and validation accuracy appeared high, the actual inference result on the XIAO was unstable. One likely reason is that the model learned shortcut features such as camera angle, lighting condition, background layout, or object placement instead of learning the true semantic difference between activities. This failure led us to replace abstract activity labels with more concrete desk-state labels that directly map to lighting behaviors.

This iteration showed that validation accuracy alone is not sufficient for an embedded vision system. A model can perform well on a validation set but still fail during live inference if the validation images are too similar to the training images or if the model relies on non-causal visual cues. Therefore, live testing on the XIAO was an important part of the design process.

4 Hardware and Lighting Control

The hardware converts the model output into visible lighting behavior while keeping the LED load separated from the microcontroller. The XIAO ESP32S3 Sense runs the classifier and produces two PWM signals for a 12 V cool-white/warm-white LED strip. The LED strip is powered by an external DC supply rather than by the XIAO board.

The system uses two PWM-capable GPIO pins, two D4184 MOSFET switching channels, and a shared ground reference. In the prototype, D1 controls the cool-white channel and D0 controls the warm-white channel. D4184 modules are commonly used for low-side DC load switching and PWM control of loads such as LED strips [16]. The MOSFET modules allow the XIAO to provide only control signals while the LED current flows through the external power path. The MOSFET device family is designed for low-resistance switching applications [17].

The ESP32S3 LEDC peripheral generates the PWM waveforms. Espressif's LEDC peripheral supports configurable PWM frequency and duty cycle generation for LED control [15]. The prototype uses 5 kHz PWM and 8-bit resolution, giving a simple 0–255 duty-cycle range. Since the camera code also uses LEDC resources, the lighting outputs use LEDC channels 2 and 3 rather than channel 0.

The key wiring requirement is a common ground between the XIAO, MOSFET modules, and the 12 V power supply. Without a shared reference, the PWM input may float and the LED brightness may become unstable. The LED strip should never be powered

Table 1: PWM and lighting-control configuration.

Signal	XIAO pin	LEDC channel	Purpose
CW	D1	2	Cool-white dimming
WW	D0	3	Warm-white dimming
GND	GND	–	Common reference
Supply	–	–	External 12 V LED power

directly from a XIAO GPIO pin; the GPIO pins only provide control signals.

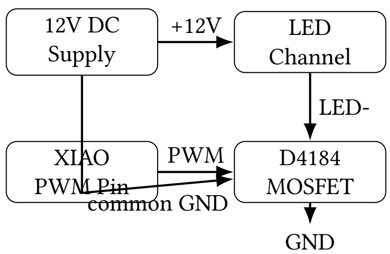

Figure 2: Low-side switching concept for one LED channel. The same structure is duplicated for CW and WW channels.

The lighting behavior is implemented with a fixed rule-based mapping from model labels to PWM duty cycles. This is simpler than calibrated color-temperature control and is sufficient for demonstrating distinct lighting modes. Table 2 shows the duty-cycle values used in the prototype.

Table 2: Model-label-to-lighting mapping used in the prototype.

Model label	CW duty	WW duty	Expected effect
computer	180/255	50/255	Cooler work light
empty	0/255	0/255	Light off
ipad	190/255	190/255	Bright reading light
phone	20/255	80/255	Dim warm light

The mapping is empirical rather than optically calibrated. The computer mode uses stronger cool-white output, the iPad mode uses a brighter mixed output, the phone mode uses a lower warm output, and the empty state turns both channels off. A future version could add a light sensor or color meter for calibration.

5 Quantization

We compared float32 and int8 versions of our MobileNetV2-based classifier exported from Edge Impulse. TensorFlow Lite Micro is designed to run machine learning models on resource-constrained embedded systems [12], and quantization is a common technique for reducing model size and enabling integer arithmetic inference [13]. The main motivation was to reduce the memory and flash footprint of the model so that it could fit more comfortably on the XIAO ESP32S3 Sense together with the camera and lighting-control code.

Table 3: Float32 and int8 deployment results on the XIAO ESP32S3 Sense.

Version	Flash (B)	RAM (B)	Cls. (ms)
Float32	2060421	34644	6228
Int8 quantized	925305	36728	6113

Table 3 shows the measured compile-time memory information and inference timing printed by the Arduino serial monitor. Quantization reduced the program storage usage from 2060421 bytes to 925305 bytes, which is more than a 50% reduction in compiled sketch size. However, the measured classification latency changed only slightly, from 6228 ms to 6113 ms.

The result shows that quantization was very effective for reducing flash usage, but it did not significantly improve inference latency in our setup. This is because quantization reduces the size of model weights, but speedup also depends on whether the deployed runtime uses optimized int8 kernels. ESP-NN provides optimized neural network functions for Espressif chips and supports TensorFlow Lite Micro, including ESP32S3-specific optimized implementations [14]. If the model runs through a more generic kernel path, the model can become much smaller without becoming much faster.

This experiment was still useful because embedded deployment is constrained by both memory and latency. Even though the speed improvement was only about 2%, the smaller int8 model left more program storage available for the camera code, Edge Impulse runtime, and PWM lighting-control firmware.

6 Discussion

The main challenge was defining a recognition task that was realistic for the XIAO ESP32S3 Sense. Fine-grained hand-object interaction recognition was too unstable for the available camera quality, dataset size, and model capacity. The first three-class model also showed that high validation accuracy can be misleading when a small fixed-view dataset allows the model to learn shortcuts. Simplifying the task to four concrete desk-state labels made the system more reliable and easier to demonstrate.

The hardware design is intentionally simple. The XIAO runs the classifier and produces two PWM signals, while the D4184 modules handle power switching. This keeps the microcontroller in the low-power logic domain and leaves the LED current path to the external 12 V supply. The main limitation is that the lighting output is not optically calibrated, and the system assumes a two-channel CW/WW LED strip. However, the prototype still provides a clear control path: image capture, on-device classification, rule-based decision making, PWM generation, and LED dimming.

7 Conclusion

This report presented a camera-based embedded AI desk lighting controller using the XIAO ESP32S3 Sense. The final prototype uses a MobileNetV2 96×96 0.35 image classification model, trained in Edge Impulse, to recognize four fixed-view desk states: computer, empty, iPad, and phone. The model achieved 93.8% validation accuracy, with the main confusion occurring between the iPad and phone classes. The predicted state is mapped to two PWM duty cycles that control cool-white and warm-white LED channels through D4184 MOSFET switching modules.

Compared with the original proposal, the final implementation narrowed the task from broad multimodal user-state monitoring to camera-only desk scene classification. This simplification was necessary because fine-grained hand-object interaction recognition was unreliable with the available camera quality, dataset size, and embedded model capacity. The quantized int8 model reduced compiled program storage by more than 50%, although inference latency improved only slightly in our test. Overall, the system demonstrates a complete embedded AI pipeline: local image capture, on-device classification, rule-based decision making, and PWM lighting control.

References

[1] Y. Fan and C. Gao. *State Monitoring and Light Control Using XIAO ESP32C3*. Course project proposal, 2026.

[2] Seeed Studio. *Getting Started with Seeed Studio XIAO ESP32S3 Series*. Seeed Studio Wiki, n.d. https://wiki.seeedstudio.com/xiao_esp32s3_getting_started/

[3] Seeed Studio. *Camera Usage in Seeed Studio XIAO ESP32S3 Sense*. Seeed Studio Wiki, 2023. https://wiki.seeedstudio.com/xiao_esp32s3_camera_usage/

[4] Edge Impulse. *Seeed XIAO ESP32S3 Sense*. Edge Impulse Documentation, n.d. https://docs.edgeimpulse.com/hardware/boards/seeed-xiao-esp32s3-sense

[5] Edge Impulse. *Image Classification*. Edge Impulse Documentation, n.d. https://docs.edgeimpulse.com/tutorials/end-to-end/image-classification

[6] Edge Impulse. *Run Arduino Library*. Edge Impulse Documentation, n.d. https://docs.edgeimpulse.com/hardware/deployments/run-arduino-2-0

[7] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen. MobileNetV2: Inverted Residuals and Linear Bottlenecks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018.

[8] A. G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, and H. Adam. MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications. arXiv:1704.04861, 2017.

[9] Edge Impulse. *FOMO: Faster Objects, More Objects*. Edge Impulse Documentation, n.d. https://docs.edgeimpulse.com/studio/projects/learning-blocks/blocks/object-detection/fomo

[10] M. Tan and Q. V. Le. EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. In *Proceedings of the International Conference on Machine Learning*, 2019.

[11] F. N. Iandola, S. Han, M. W. Moskewicz, K. Ashraf, W. J. Dally, and K. Keutzer. SqueezeNet: AlexNet-Level Accuracy with 50x Fewer Parameters and < 0.5MB Model Size. arXiv:1602.07360, 2016.

[12] R. David, J. Duke, A. Jain, V. J. Reddi, N. Jeffries, J. Li, N. Kreeger, I. Nappier, M. Natraj, S. Regev, R. Rhodes, T. Wang, and P. Warden. TensorFlow Lite Micro: Embedded Machine Learning on TinyML Systems. In *Proceedings of Machine Learning and Systems*, 2021.

[13] B. Jacob, S. Kligys, B. Chen, M. Zhu, M. Tang, A. Howard, H. Adam, and D. Kalenichenko. Quantization and Training of Neural Networks for Efficient Integer-Arithmetic-Only Inference. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018.

[14] Espressif Systems. *ESP-NN: Optimized Neural Network Functions for Espressif Chips*. GitHub repository, n.d. https://github.com/espressif/esp-nn

[15] Espressif Systems. *LED Control (LEDC) API for Arduino ESP32*. Espressif Documentation, n.d.

[16] ProtoSupplies. *D4184 MOSFET Control Module*. Product documentation, n.d. https://protosupplies.com/product/d4184-mosfet-control-module/

[17] Alpha & Omega Semiconductor. *AOD4184A 40V N-Channel MOSFET Datasheet*. Product datasheet, n.d.

[18] D. Caicedo and A. Pandharipande. Occupancy-based illumination control of LED lighting systems. *Lighting Research & Technology*, 45(3), 344–358, 2013.

[19] Y. Agarwal, B. Balaji, S. Dutta, R. K. Gupta, and T. Weng. Occupancy-driven energy management for smart building automation. In *Proceedings of ACM BuildSys*, 2010.